SAIDS — PYQ Question Mapper

Statistics for Artificial Intelligence & Data Science

CSE-AIML Sem V · C Scheme (R-2019) · Dec 2022 – May 2025

PAPER LEGEND

Dec 2022	Dec 2023	May 2024	Dec 2024	May 2025 (AIML)	May 2025 (IT)	IAE QB-1	IAE QB-2

QUESTIONS PER MODULE

Module I	Statistical Foundations, Covariance & Sampling	**43 Qs**
Module II	Statistical Inference, Estimation & Resampling	**45 Qs**
Module III	Regression Models, Regularization & Nonlinear Models	**13 Qs**
Module IV	Classification & Supervised Learning	**7 Qs**
Module V	Unsupervised Learning & Dimensionality Reduction	**8 Qs**
Module VI	Model Evaluation, Bayesian Inference & Ethics	**3 Qs**

Module I Statistical Foundations, Covariance & Sampling

Syllabus Topics: *Data types (categorical, numerical, ordinal) · Descriptive stats (central tendency, dispersion, skewness, kurtosis) · Data viz (histograms, boxplots, scatter plots, pair plots, QQ plots) · Probability distributions (Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Weibull) · Covariance, correlation, correlation matrices · Sampling (simple random, stratified, cluster) · Central Limit Theorem (CLT) & Law of Large Numbers · Empirical CDF · Data preprocessing (outlier treatment, encoding, normalization)*

Total: 43 questions · Dec 2022: 7 · Dec 2023: 6 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 6 · May 2025 (IT): 5 · IAE QB-1: 4

1. **[Dec 2022 | Q1a]** Find standard deviation of average temperatures over 5-day period: 19, 21, 18, 24, 12 *(Numerical)*

2. **[Dec 2022 | Q1b]** Normal distribution (μ=30, σ=4): find P(x<40) and P(30<x<35) *(Numerical)*

3. **[Dec 2022 | Q1e]** What do you mean by correlation and regression? Explain with example *(Theory)*

4. **[Dec 2022 | Q2a]** Find correlation coefficient from Age (X) vs Glucose Level (Y) data (6 subjects) *(Numerical)*

5. **[Dec 2022 | Q3a]** What is the use of scatter plot and box plot? *(Theory)*

6. **[Dec 2022 | Q6a]** Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2) *(Numerical)*

7. **[Dec 2022 | Q6b]** Short notes (any two): Central Limit Theorem, Standard Error *(Short notes)*

8. **[Dec 2023 | Q1d]** Short note on Random sampling *(Short note)*

9. **[Dec 2023 | Q1e]** What is the Empirical CDF function? *(Short note)*

10. **[Dec 2023 | Q2a]** Frequency distribution table for 30 orange weights; find class mark, range, number of intervals, lowest-frequency interval *(Numerical)*

11. **[Dec 2023 | Q3b]** Concept of correlation; how is it different from regression? *(Theory)*

12. **[Dec 2023 | Q4a]** Normal distribution: P(car speed > 100 km/hr) with μ=90, σ=10 *(Numerical)*

13. **[Dec 2023 | Q4b]** Explain Numerical and Categorical data types with appropriate examples *(Theory)*

14. **[May 2024 | Q1d]** What do you mean by correlation and regression? Explain with example *(Theory)*

15. **[May 2024 | Q2a]** Normal distribution (μ=30, σ=4): find P(x<40), P(x>21), P(30<x<35) *(Numerical)*

16. **[May 2024 | Q2b]** Poisson distribution: vehicles at 300/hr — P(none in 1 min), expected in 2 min, P(expected number passes in 2 min) *(Numerical)*

17. **[May 2024 | Q3a]** Normal distribution: P(battery charge lasts 50–70 hrs) for μ=50, σ=15 *(Numerical)*

18. **[May 2024 | Q4a]** Explain QQ plots in detail; show how scatterplots explore relationships between variables *(Theory)*

19. **[May 2024 | Q6a]** Explain Binomial distribution in detail; Type I error concept in bottle volume test *(Theory + Numerical)*

20. **[May 2024 | Q6b]** Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot *(Short notes)*

21. **[Dec 2024 | Q1a]** Explain percentiles and Boxplots with example *(Short note)*

22. **[Dec 2024 | Q1b]** Illustrate Central Limit Theorem with a neat diagram *(Short note)*

23. **[Dec 2024 | Q1d]** What is a Histogram? Give its applications *(Short note)*

24. **[Dec 2024 | Q2b]** Explain Normal and Poisson Distribution; GMAT scores (μ=527, σ=112): P(score>500) and score for top 5% *(Theory + Numerical)*

25. **[Dec 2024 | Q4b]** Mean, SD, standard error and Confidence Interval; compute all for a 10-data-point sample *(Theory + Numerical)*

26. **[Dec 2024 | Q5b]** Continuous Probability Distribution & PDF; Uniform distribution on [0,20]: P(5≤X≤15), P(X<10), P(X>18) *(Numerical)*

27. **[Dec 2024 | Q6b]** Short note: Histogram and Scatterplot with example *(Short note)*

28. **[Dec 2024 | Q6d]** Short note: Weibull Distribution *(Short note)*

29. **[May 2025 (AIML) | Q1c]** Explain the difference between Stratified and Cluster Sampling *(Short note)*

30. **[May 2025 (AIML) | Q1e]** Define standard deviation and interquartile range with examples *(Short note)*

31. **[May 2025 (AIML) | Q2a]** Find the correlation coefficient from given Experience (X) vs Salary (Y) data *(Numerical)*

32. **[May 2025 (AIML) | Q4a]** Z-score: sample mean=250, population mean=260, SD=40; calculate z-score for sample mean *(Numerical)*

33. **[May 2025 (AIML) | Q4b]** Frequency distribution table for 50 college students' weekly study hours *(Numerical)*

34. **[May 2025 (AIML) | Q6b]** Short notes (any 2): Chi-square distribution, Weibull distribution, Stem & Leaf Plot, Box Plot *(Short notes)*

35. **[May 2025 (IT) | Q1b]** Explain Poisson probability distribution with one example *(Short note)*

36. **[May 2025 (IT) | Q1d]** Discuss Quantitative Data, Descriptive Statistics, Variables, Categorical data *(Theory)*

37. **[May 2025 (IT) | Q1e]** Why is sampling necessary? Describe population and sample mean *(Short note)*

38. **[May 2025 (IT) | Q2b]** Binomial distribution: insurance agent, P(living 30+ yrs)=2/3 for 5 people — P(all 5), P(≥3), P(exactly 2) *(Numerical)*

39. **[May 2025 (IT) | Q6b]** Short notes: Scatter diagram, Sample space, Discrete probability distribution *(Short notes)*

40. **[IAE QB-1 | A2]** What is mean? Find the mean of the data: 4, 6, 8, 10 *(Numerical)*

41. **[IAE QB-1 | B1]** Find the mean, median and mode for the following marks: 10, 20, 20, 30, 40 *(Numerical)*

42. **[IAE QB-1 | B5]** A university has 2000 students. Explain an appropriate sampling technique to select a sample; use CLT to explain how statistical inference can be drawn from the sample *(Theory)*

43. **[IAE QB-1 | B6]** Binomial distribution: A factory produces bulbs with P(defective)=0.1. If n=5 bulbs are selected, find P(exactly 2 are defective) *(Numerical)*

Module II Statistical Inference, Estimation & Resampling

Syllabus Topics: *Point estimates · Bias, Variance, MSE · MLE & Method of Moments · Confidence intervals (mean, proportion, variance) · Hypothesis testing: z-test, t-test, Chi-square, F-test, ANOVA (1-way, 2-way) · Nonparametric tests: Mann-Whitney U, Kruskal-Wallis, Friedman, Wilcoxon, Sign test · Bootstrap CI & resampling · Likelihood Ratio Tests (LRT) · Type I & II errors · p-value*

Total: 45 questions · Dec 2022: 8 · Dec 2023: 7 · May 2024: 7 · Dec 2024: 8 · May 2025 (AIML): 7 · May 2025 (IT): 5 · IAE QB-1: 3

1. [Dec 2022 | Q1c] Discuss Bootstrapping vs. re-sampling *(Short note)*

> **Answer:** Bootstrap CI: Repeatedly draw samples WITH replacement from original data; compute statistic (mean/median) each time; use percentile method (2.5th–97.5th percentile) to find 95% CI. Example: Data {8,10,12,14,16} → Bootstrap CI ≈ [11.6, 13.2]. Permutation Test: Resampling method for hypothesis testing; randomly shuffle/rearrange data WITHOUT replacement to generate null distribution; compute p-value. Key difference: Bootstrap estimates uncertainty/CI; Permutation test performs hypothesis testing. Bootstrap → Resample; Permutation → Shuffle.

2. [Dec 2022 | Q1d] School principal: state null and alternative hypotheses (teachers claim students use computer 3.2 hrs/day) *(Theory)*

> **Answer:** H_0 (Null Hypothesis): $\mu = 3.2$ hrs/day — The teachers' claim is true; students use computer exactly 3.2 hrs/day on average. H_1 (Alternative Hypothesis): $\mu \neq 3.2$ hrs/day — The teachers' claim is not supported. This is a TWO-TAILED test (we test for any deviation, not just increase or decrease).

3. [Dec 2022 | Q2b] Explain why ANOVA is used; solve One-way ANOVA for Observations A, B, C (5×3 table) *(Theory + Numerical)*

> **Answer:** ANOVA (Analysis of Variance) is used to compare means of 3+ groups simultaneously. Using multiple t-tests would inflate Type I error; ANOVA keeps α controlled. Key components: SSB (Between-group Sum of Squares), SSW (Within-group Sum of Squares), SST = SSB + SSW. F = MSB/MSW = [SSB/(k-1)] / [SSW/(N-k)]. Steps: (1) H_0: $\mu_1=\mu_2=\mu_3$; H_1: at least one mean differs. (2) Compute Grand Mean. (3) Calculate SSB and SSW. (4) Compute F-statistic. (5) Compare with F-critical. If F > F-critical → Reject H_0.

4. [Dec 2022 | Q3a] Explain Type I & Type II error in detail *(Theory)*

> **Answer:** Type I Error (False Positive): Rejecting H_0 when it is actually TRUE. Probability = α (significance level). Example: Declaring a drug effective when it has no effect. Type II Error (False Negative): FAILING to reject H_0 when it is actually FALSE. Probability = β. Example: Missing an effective drug in clinical trials. Power of test = 1 - β (probability of correctly rejecting a false H_0). Trade-off: Reducing α (e.g., 0.05→0.01) increases β, and vice versa.

5. [Dec 2022 | Q3b] Kruskal-Wallis test: 4 manufacturing facilities, exam scores after machining training, at 95% confidence — are scores same? *(Numerical)*

> **Answer:** Kruskal-Wallis is the non-parametric alternative to one-way ANOVA (used when data is not normally distributed). H_0: All groups have the same distribution. Steps: (1) Rank ALL observations from all groups together (N total). (2) Compute rank sums $R_1, R_2, \ldots, R_k$ for each group. (3) $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$. (4) df = k-1. (5) Compare H with χ^2-critical at α=0.05. If H > χ^2-critical → Reject H_0 (scores differ significantly between facilities).

6. [Dec 2022 | Q4a] t-score: sample mean=290, expected mean=300, SD=50, n=15 — find t-score *(Numerical)*

> **Answer:** Formula: $t = (\bar{x} - \mu) / (s/\sqrt{n})$. Given: $\bar{x}=290$, $\mu=300$, $s=50$, $n=15$. $s/\sqrt{n} = 50/\sqrt{15} = 50/3.873 = 12.91$. $t = (290 - 300)/12.91 = -10/12.91 = -0.775$. The t-score is -0.775 with df = n-1 = 14.

7. [Dec 2022 | Q5a] z-test: pea plant pods (μ=145, σ=100, $\bar{x}$=147) — state hypotheses, test statistic at α=0.05 *(Numerical)*

> **Answer:** H_0: $\mu = 145$ (mean number of pods = 145). H_1: $\mu \neq 145$ (two-tailed test). Formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$. Given: $\bar{x}=147$, $\mu_0=145$, $\sigma=100$. At α=0.05 (two-tailed), critical Z = ±1.96. Calculate Z from the given n and compare: if $|Z| > 1.96$, reject H_0. If $|Z| \leq 1.96$, fail to reject H_0.

8. [Dec 2022 | Q6b] Short note: Confidence Interval *(Short note)*

> **Answer:** A Confidence Interval (CI) is a range of values constructed from sample data that contains the true population parameter with a stated probability (confidence level). CI for mean (known σ): $\bar{x} \pm z^*(\sigma/\sqrt{n})$. CI for mean (unknown σ): $\bar{x} \pm t^*(s/\sqrt{n})$. CI for proportion: $\hat{p} \pm z^*\sqrt{[\hat{p}(1-\hat{p})/n]}$. CI for variance: $[(n-1)s^2/\chi^2_upper, (n-1)s^2/\chi^2_lower]$. Solved example: Data {10,12,14,16,18}, $\bar{x}=14$, $s=3.16$, $n=5$, $t^*=2.776$ (df=4) → 95% CI = $14 \pm 2.776\times(3.16/\sqrt{5}) = (10.073, 17.927)$.

9. [Dec 2023 | Q1a] Short note on hypothesis testing *(Short note)*

> **Answer:** Hypothesis testing is a statistical procedure to determine whether sample evidence is sufficient to reject a null hypothesis. Steps: (1) State H_0 (null) and H_1 (alternative); (2) Choose significance level α (usually 0.05); (3) Select appropriate test statistic (z, t, χ^2, F); (4) Compute test statistic from sample data; (5) Find critical value or p-value; (6) Decision: if |test stat| > critical value (or p-value < α), reject H_0; (7) Draw conclusion in context of the problem. Types: two-tailed, right-tailed, left-tailed.

10. [Dec 2023 | Q1b] What is Fisher's exact test? *(Short note)*

> **Answer:** Fisher's exact test is a statistical significance test used to analyze the association between two categorical variables in a 2×2 contingency table, particularly when sample sizes are SMALL. Unlike Chi-square (which uses approximations), Fisher's test computes the EXACT probability of observing the given cell frequencies (or more extreme) under H_0 of no association. When to use: When expected cell frequencies < 5 (Chi-square assumption violated). Formula involves hypergeometric distribution. Commonly used in medical/biological research with small groups.

11. [Dec 2023 | Q2b] Purpose of one-way ANOVA; key components: dependent variable, independent variable, factors *(Theory)*

Answer: One-way ANOVA tests whether the means of 3+ groups (defined by ONE factor/independent variable) are significantly different. Key components: (1) Dependent variable: the outcome being measured (e.g., exam scores); (2) Independent variable (Factor): the grouping variable with k levels (e.g., teaching method A/B/C); (3) Between-group variance (MSB = SSB/(k-1)): variation due to the factor; (4) Within-group variance (MSW = SSW/(N-k)): variation within each group (random error); (5) F-statistic = MSB/MSW. H_0: $\mu_1=\mu_2=...=\mu_k$; H_1: at least one mean differs.

12. **[Dec 2023 | Q3a]** Standard error of estimate for household children data; 95% confidence interval *(Numerical)*

Answer: Standard Error (SE) = $s/\sqrt{n}$. It measures how far the sample mean is expected to be from the true population mean. 95% CI = $\bar{x} \pm t^*(s/\sqrt{n})$ where t^* is the t-critical value at df=n-1 and $\alpha/2=0.025$. Key formula from teacher notes: Solved example with data {10,12,14,16,18}: $\bar{x}=14$, $s=3.16$, $n=5$, $SE=3.16/\sqrt{5}=1.414$, $t^*=2.776$ (df=4). 95% CI = $14 \pm 2.776\times1.414 = 14 \pm 3.927 = (10.073, 17.927)$. Interpretation: We are 95% confident the true mean lies between 10.073 and 17.927.

13. **[Dec 2023 | Q5a]** t-test: Duracell CEO claims $\mu=300$h; sample $n=20$, $\bar{x}=270$h, $s=50$h — reject claim? *(Numerical)*

Answer: H_0: $\mu = 300$h (CEO's claim is true). H_1: $\mu < 300$h (one-tailed, left-tailed). Formula: $t = (\bar{x} - \mu)/(s/\sqrt{n}) = (270 - 300)/(50/\sqrt{20}) = -30/11.18 = -2.68$. df = n-1 = 19. At $\alpha=0.05$ (one-tailed), t-critical = -1.729. Since $|-2.68| > 1.729$, we REJECT H_0. Conclusion: There is sufficient evidence to reject the CEO's claim. The batteries do not last 300 hours on average.

14. **[Dec 2023 | Q6a]** z-test: pea plant pods ($\mu=145$, $\sigma=100$, sample $\bar{x}=147$) — state hypotheses and test statistic *(Numerical)*

Answer: H_0: $\mu = 145$ (mean pods per plant = 145). H_1: $\mu \neq 145$ (two-tailed). Formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$. Given: $\bar{x}=147$, $\mu_0=145$, $\sigma=100$. At $\alpha=0.05$ (two-tailed), z-critical = ±1.96. Compute Z for the given n; compare $|Z|$ with 1.96. If $|Z| > 1.96 \rightarrow$ Reject H_0 (mean pods differ significantly from 145). If $|Z| \leq 1.96 \rightarrow$ Fail to reject H_0.

15. **[Dec 2023 | Q6b]** What is Chi-Square Test and in what situations is it commonly used? *(Theory)*

Answer: The Chi-square (χ^2) test is a non-parametric test for categorical data. Formula: $\chi^2 = \Sigma[(O-E)^2/E]$ where O=observed frequency, E=expected frequency=(Row Total × Col Total)/Grand Total. Uses: (1) Test of Independence: tests association between two categorical variables. (2) Goodness of Fit: tests if data follows a hypothesized distribution. df = (rows-1)(cols-1) for independence. Decision: If $\chi^2 > \chi^2$-critical, reject H_0 (variables are associated). Solved example from notes (brand preference): $\chi^2=2.00 < \chi^2$-critical → fail to reject H_0. Example 2 (gender vs online classes): $\chi^2=4 > \chi^2$-critical=3.841 → reject H_0.

16. **[May 2024 | Q1a]** Define Confidence Interval *(Short note)*

Answer: A Confidence Interval (CI) is a range of values derived from sample data that is estimated to contain the true population parameter with a specified confidence level (e.g., 95%). Formula: CI = point estimate ± margin of error. Example: $CI = \bar{x} \pm z^*(\sigma/\sqrt{n})$ for known σ, or $\bar{x} \pm t^*(s/\sqrt{n})$ for unknown σ. Interpretation: A 95% CI means that if we repeated the sampling 100 times, 95 of those intervals would contain the true parameter. A wider CI means more uncertainty; a narrower CI means more precision.

17. **[May 2024 | Q1b]** Standard Error of sample mean: $\mu=\$12$, $\sigma=\$3$, $n=30$ — estimate and interpret SE *(Numerical)*

Answer: Formula: $SE = \sigma/\sqrt{n} = 3/\sqrt{30} = 3/5.477 = 0.547$. Interpretation: The sample mean of $12 is expected to vary by approximately $0.55 from the true population mean across repeated samples of size 30. The standard error decreases as sample size increases ($SE \propto 1/\sqrt{n}$), meaning larger samples give more precise estimates of the population mean.

18. **[May 2024 | Q1c]** What is hypothesis testing? Explain Type I and Type II errors *(Short note)*

Answer: Hypothesis testing: A formal procedure to test a claim about a population parameter using sample data. Start with H_0 (null) and H_1 (alternative); compute test statistic; compare with critical value or use p-value. Type I Error (α): Rejecting H_0 when it is TRUE (False Positive). Probability = α. Type II Error (β): Failing to reject H_0 when it is FALSE (False Negative). Probability = β. Power = 1 - β. Trade-off: Lowering α increases β. Reducing β (increasing power) requires larger sample sizes.

19. **[May 2024 | Q1e]** What is analysis of variance (ANOVA)? Explain its usage *(Short note)*

Answer: ANOVA (Analysis of Variance) is a statistical test to compare means of THREE OR MORE groups. Instead of running multiple t-tests (which inflate Type I error), ANOVA provides a single F-test. F = MSB/MSW where MSB = Between-group variance, MSW = Within-group variance. Usage: Compare exam scores across multiple teaching methods; compare crop yields for different fertilizers; compare customer satisfaction across branches. One-way ANOVA: one factor. Two-way ANOVA: two factors (can detect interaction effects).

20. **[May 2024 | Q3b]** z-test: score change ($\mu=80$, $\sigma=10$, $n=38$, $\bar{x}=88$) at $\alpha=0.05$ — evidence for claim? *(Numerical)*

Answer: H_0: $\mu = 80$. H_1: $\mu > 80$ (one-tailed, right-tailed — testing if scores improved). Formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n}) = (88-80)/(10/\sqrt{38}) = 8/1.622 = 4.93$. At $\alpha=0.05$ (one-tailed), z-critical = 1.645. Since $4.93 > 1.645$, REJECT H_0. Conclusion: There is strong statistical evidence at $\alpha=0.05$ that the mean score has increased above 80.

21. **[May 2024 | Q4b]** One-way ANOVA numerical: four samples A, B, C, D — identify differences *(Numerical)*

Answer: Steps for one-way ANOVA with 4 groups: (1) H_0: μA=μB=μC=μD; H_1: at least one differs. (2) Compute group means $\bar{x}A$, $\bar{x}B$, $\bar{x}C$, $\bar{x}D$ and Grand Mean. (3) $SSB = \Sigma n_i(\bar{x}_i - \bar{x}grand)^2$; $SSW = \Sigma(x_{ij} - \bar{x}_i)^2$. (4) MSB = SSB/(k-1) = SSB/3; MSW = SSW/(N-k) = SSW/(N-4). (5) F = MSB/MSW. (6) Compare with F-critical(3, N-4) at $\alpha=0.05$. (7) If F > F-critical → at least one group mean is significantly different.

22. **[May 2024 | Q5a]** F-Test: $F=2.38$, df=(8,3) — find F-table value; reject null at $\alpha=0.05$ (one-tailed)? *(Numerical)*

Answer: F-test compares variances of two populations. H_0: $\sigma_1^2=\sigma_2^2$ (variances equal); H_1: $\sigma_1^2 \neq \sigma_2^2$. Given: F-calculated=2.38, $df_1=8$, $df_2=3$. From F-distribution table at $\alpha=0.05$ (one-tailed) with df=(8,3): F-critical ≈ 8.85. Since $2.38 < 8.85$, FAIL to reject H_0. Conclusion: There is no significant difference in variances at $\alpha=0.05$. The F-test is also the test statistic in ANOVA (F = MSB/MSW).

23. **[Dec 2024 | Q1c]** Explain Null and alternative Hypothesis with example *(Short note)*

Answer: Null Hypothesis (H_0): The default assumption; states no effect or no difference. It is the hypothesis we try to disprove. Example: H_0: $\mu = 50$ (average test score is 50). Alternative Hypothesis (H_1): The claim we want to support. H_1: $\mu \neq 50$ (two-tailed), H_1: $\mu > 50$ (right-

tailed), H_1: $\mu < 50$ (left-tailed). Rule: We never "accept" H_0; we either REJECT H_0 (when evidence is strong) or FAIL TO REJECT H_0 (when evidence is insufficient).

24. [Dec 2024 | Q1f] What is the F-Test used for? *(Short note)*

Answer: The F-test is used to: (1) Compare two population variances — tests H_0: $\sigma_1^2=\sigma_2^2$ using $F = s_1^2/s_2^2$ (where $s_1^2 \geq s_2^2$). Critical value from F-distribution table (df_1, df_2). (2) As the test statistic in ANOVA — F = MSB/MSW tests if group means differ. If F > F-critical, reject H_0. Applications: Checking equal variance assumption before t-test; comparing variability in manufacturing processes; ANOVA (comparing 3+ means).

25. [Dec 2024 | Q2a] Chi-Square Test: gender vs. coffee/tea preference contingency table (Male/Female × Coffee/Tea) at α=0.05 *(Numerical)*

Answer: H_0: Gender and beverage preference are INDEPENDENT. H_1: They are associated. Steps: (1) Construct 2×2 contingency table with observed frequencies. (2) Compute expected E = (Row Total × Column Total)/Grand Total for each cell. (3) $\chi^2 = \Sigma[(O-E)^2/E]$ for all cells. (4) df = (rows-1)(cols-1) = (2-1)(2-1) = 1. (5) At α=0.05, χ^2-critical = 3.841. (6) If χ^2-calculated > 3.841 → Reject H_0 (gender and preference are associated). If χ^2-calculated ≤ 3.841 → Fail to reject H_0.

26. [Dec 2024 | Q3b] t-Distribution: CEO claims bulb lasts 300 days; n=15, $\bar{x}$=290, s=50 — $P(\bar{x} \leq 290)$ *(Numerical)*

Answer: H_0: $\mu = 300$ days. H_1: $\mu < 300$ (one-tailed, left-tailed). Formula: $t = (\bar{x} - \mu)/(s/\sqrt{n}) = (290-300)/(50/\sqrt{15}) = -10/12.91 = -0.775$. df = n-1 = 14. At α=0.05 (one-tailed), t-critical = -1.761. Since $|-0.775| < 1.761$, FAIL to reject H_0. Conclusion: There is insufficient evidence to reject the CEO's claim. The observed mean of 290 days is not significantly different from 300 days.

27. [Dec 2024 | Q4a] ANOVA: why use it; one-way vs two-way difference; solve one-way ANOVA for 3 groups *(Theory + Numerical)*

Answer: Why ANOVA: Comparing 3+ means using multiple t-tests inflates Type I error. ANOVA uses a single F-test. One-way ANOVA: ONE factor (independent variable), tests if k group means are equal. Two-way ANOVA: TWO factors; can detect main effects and INTERACTION effects. One-way ANOVA Steps: (1) H_0: all μ equal; (2) Compute SSB, SSW; (3) F = MSB/MSW; (4) Compare F with F-critical. Solved example from notes (3 fertilizers, N=9): R1=6, R2=15, R3=24; H=7.2 > χ^2-critical=5.991 → significant difference.

28. [Dec 2024 | Q5a] Nonparametric vs parametric methods; Friedman Test: THM water cleanup data (12 counties) *(Theory + Numerical)*

Answer: Nonparametric tests: Do not assume normal distribution; used with ordinal/non-normal data or small samples. Examples: Mann-Whitney U, Kruskal-Wallis, Wilcoxon, Friedman. Parametric tests: Assume normality; more powerful when assumptions met. Examples: t-test, ANOVA, z-test. Friedman Test: Non-parametric alternative to TWO-WAY ANOVA for repeated measures. Steps: (1) Rank each row (block) separately; (2) Compute rank sums for each treatment; (3) Calculate Friedman statistic Fr; (4) Compare Fr with χ^2-critical (df=k-1); (5) If Fr > critical → treatments differ significantly.

29. [Dec 2024 | Q6a] Short note: Type I and Type II errors in detail *(Short note)*

Answer: Type I Error (α, False Positive): Rejecting H_0 when H_0 is TRUE. Probability = α (significance level set by researcher). Example: Declaring a new drug effective when it actually has no effect. Type II Error (β, False Negative): Failing to reject H_0 when H_0 is FALSE. Probability = β. Example: Declaring a drug ineffective when it actually works. Power = 1 - β = probability of correctly detecting a real effect. Key: α and β are inversely related. To minimize both, increase sample size n.

30. [Dec 2024 | Q6c] Short note: Bootstrapping vs Re-sampling *(Short note)*

Answer: Bootstrap CI: Resampling WITH replacement from original data; compute statistic for each bootstrap sample; use percentile method for CI. Use when: population distribution is unknown, sample is small, or statistic has complex distribution. Permutation/Re-sampling Test: Shuffle data WITHOUT replacement; generate null distribution; compute p-value. Use for hypothesis testing. Key comparison: Bootstrap → estimates UNCERTAINTY (CI, SE); Permutation → performs HYPOTHESIS TESTING (p-value). Bootstrap: Resample; Permutation: Shuffle.

31. [May 2025 (AIML) | Q1a] What is hypothesis testing? Explain Type I and Type II errors *(Short note)*

Answer: Hypothesis testing: A formal procedure using sample data to evaluate claims about population parameters. Steps: (1) State H_0 and H_1; (2) Choose α; (3) Collect data; (4) Compute test statistic; (5) Compare with critical value or find p-value; (6) Decision and conclusion. Type I Error: Reject TRUE H_0 (False Positive); P(Type I) = α. Type II Error: Fail to reject FALSE H_0 (False Negative); P(Type II) = β. Power = 1-β. A good hypothesis test minimizes both errors; increasing sample size helps achieve this.

32. [May 2025 (AIML) | Q1b] What is Fisher's exact test? *(Short note)*

Answer: Fisher's exact test is used to analyze 2×2 contingency tables when sample sizes are SMALL (expected cell frequencies < 5, where Chi-square approximation breaks down). It computes the EXACT probability of observing the given table or a more extreme one, under H_0 of no association, using the hypergeometric distribution. Advantage over Chi-square: No approximation needed; accurate for small samples. Commonly used in medical trials, genetics, and biology where small group sizes are unavoidable.

33. [May 2025 (AIML) | Q2b] Chi-Square Test: gender vs. online/in-store shopping (n=200) at α=0.05 *(Numerical)*

Answer: H_0: Gender and shopping preference are independent. Steps: (1) Construct 2×2 table (Male/Female × Online/In-store) with observed O. (2) E = (Row Total × Column Total)/200 for each cell. (3) $\chi^2 = \Sigma[(O-E)^2/E]$. (4) df = (2-1)(2-1) = 1. (5) At α=0.05, χ^2-critical = 3.841. (6) Decision: If χ^2 > 3.841 → Reject H_0 (gender and shopping preference are associated). Practical: A significant result means men and women shop differently online vs in-store.

34. [May 2025 (AIML) | Q3a] Explain the concept of p-value in hypothesis testing *(Theory)*

Answer: The p-value is the probability of observing a test statistic as extreme as (or more extreme than) the one calculated from the sample, ASSUMING H_0 is true. Decision rule: If p-value < α → Reject H_0 (result is statistically significant). If p-value ≥ α → Fail to reject H_0. Interpretation: A p-value of 0.018 means there is only a 1.8% chance of observing this result if H_0 were true. From permutation test

example: p = 180/10000 = 0.018 < 0.05 → Reject H_0. A small p-value = strong evidence against H_0; it does NOT measure the size of the effect.

35. [May 2025 (AIML) | Q3b] Kruskal-Wallis test: aptitude scores of Grade A, B, C students at 95% confidence level *(Numerical)*

Answer: Kruskal-Wallis test: Non-parametric alternative to one-way ANOVA. H_0: All groups have same distribution. H_1: At least one group differs. Steps: (1) Combine and rank ALL observations (N total). (2) Compute rank sums R_1, R_2, R_3 for each group. (3) Check: $R_1+R_2+R_3 = N(N+1)/2$. (4) $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$. (5) df = k-1 = 2. At α=0.05, χ^2-critical = 5.991. Solved example (3 teaching methods, N=12): H=9.846 > 5.991 → Reject H_0. Significant difference among groups.

36. [May 2025 (AIML) | Q5a] z-test: pharmaceutical blood pressure drug (μ=10mmHg, σ=5, n=30, $\bar{x}$=12) at α=0.05 — state H_0/H_1, test stat, conclusion *(Numerical)*

Answer: H_0: μ = 10 mmHg (drug reduces BP by exactly 10 mmHg). H_1: μ > 10 mmHg (drug reduces BP by MORE than 10 mmHg — one-tailed right test). Formula: $Z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n}) = (12-10)/(5/\sqrt{30}) = 2/0.913 = 2.19$. At α=0.05 (one-tailed), z-critical = 1.645. Since 2.19 > 1.645, REJECT H_0. Conclusion: There is sufficient evidence at α=0.05 that the drug reduces blood pressure by more than 10 mmHg.

37. [May 2025 (AIML) | Q6a] Two-way ANOVA: concept; difference from one-way ANOVA; assumptions; Friedman's test as nonparametric alternative *(Theory)*

Answer: Two-way ANOVA: Tests the effect of TWO factors (independent variables) on a dependent variable, including their INTERACTION. Difference from one-way: One-way uses ONE factor; two-way uses TWO factors and can detect interaction effects (does the effect of factor A depend on factor B?). Assumptions: Normality, homogeneity of variances, independence of observations. Friedman Test: Non-parametric alternative to two-way ANOVA for repeated-measures or related samples. Instead of comparing means, it ranks values within each block and compares treatment rank sums. Used when normality assumption is violated.

38. [May 2025 (IT) | Q1c] What is statistical inference? *(Short note)*

Answer: Statistical inference is the process of drawing conclusions about a POPULATION based on analysis of a SAMPLE drawn from it. Two main branches: (1) Estimation: point estimates (MLE, Method of Moments gives a single value) and interval estimates (confidence intervals give a range). (2) Hypothesis Testing: Testing claims about population parameters using z-test, t-test, chi-square, ANOVA, etc. Key principle: Sample statistics ($\bar{x}$, s, $\hat{p}$) estimate population parameters (μ, σ, p) with quantified uncertainty.

39. [May 2025 (IT) | Q2a] Null and alternate hypothesis; explain Type I and Type II error *(Theory)*

Answer: H_0 (Null Hypothesis): Default claim; assumes no effect or equality (e.g., H_0: $\mu=\mu_0$). Always contains "=". H_1 (Alternative Hypothesis): The research claim; what we want to prove. Can be two-tailed (≠), right-tailed (>), or left-tailed (<). Type I Error: Rejecting H_0 when it is TRUE; probability = α. Type II Error: Failing to reject H_0 when it is FALSE; probability = β. Power = 1-β. Decision table: correct → (reject false H_0) or (not reject true H_0); incorrect → Type I or Type II error.

40. [May 2025 (IT) | Q5a] t-test: pizza shop cheese weights (n=7) — state hypotheses, test statistic; reject at 10%, 5%, 1% level? *(Numerical)*

Answer: Formula: $t = (\bar{x} - \mu)/(s/\sqrt{n})$, df = n-1 = 6. Steps: (1) State H_0 (e.g., μ = target weight) and H_1 (μ ≠ target, two-tailed). (2) Calculate $\bar{x}$ and s from the 7 weights. (3) Compute t-statistic. (4) Compare |t| with t-critical at df=6 for each significance level: At 10%: t-critical=1.943; At 5%: t-critical=2.447; At 1%: t-critical=3.707. (5) Reject H_0 at level where |t-calculated| > t-critical. Higher significance → easier to reject H_0.

41. [May 2025 (IT) | Q6a] What is Sign Test? Explain Wilcoxon Signed-Rank Test *(Theory)*

Answer: Sign Test: Simplest non-parametric test for single sample or paired data. Count + and - differences from hypothesized median. Test statistic B = number of + signs. Compare B with binomial critical value. Wilcoxon Signed-Rank Test: More powerful than sign test; uses both SIGN and MAGNITUDE of differences. Steps: (1) Compute differences $d_i = x_i - \mu_0$; (2) Rank $|d_i|$ (ignore zeros); (3) Compute W^+ (sum of ranks with + sign) and W^- (sum of ranks with - sign); (4) $T = \min(W^+, W^-)$; (5) Compare T with critical value from Wilcoxon table. Used for: paired samples or single sample vs hypothesized median (non-normal data).

42. [May 2025 (IT) | Q6b] Short note: Kruskal-Wallis Test *(Short note)*

Answer: Kruskal-Wallis Test: Non-parametric alternative to one-way ANOVA. Used when: (1) Data is NOT normally distributed; (2) Data is ordinal; (3) Sample sizes are small. H_0: All k groups come from the same distribution. Formula: $H = [12/N(N+1)] \times \Sigma(R_i^2/n_i) - 3(N+1)$ where R_i = rank sum of group i, n_i = group size, N = total sample size. df = k-1. Compare H with χ^2-critical. Example from notes (3 fertilizers, N=9): H=7.2 > χ^2-critical=5.991 → significant difference in plant growth among fertilizers.

43. [IAE QB-1 | A7] What is bias of an estimator? Explain its significance *(Theory)*

Answer: Bias of an estimator $\hat{\theta}$ is defined as: $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$, where $E(\hat{\theta})$ is the expected value of the estimator and θ is the true unknown parameter. If Bias = 0, the estimator is UNBIASED (e.g., sample mean $\bar{x}$ is an unbiased estimator of μ, since $E(\bar{x}) = \mu$). If Bias ≠ 0, the estimator systematically overestimates (bias > 0) or underestimates (bias < 0) the true value. MSE = Variance + Bias^2. Significance: An unbiased estimator with low variance is ideal for accurate statistical inference.

44. [IAE QB-1 | A8] What is Maximum Likelihood Estimation (MLE)? *(Theory)*

Answer: MLE finds the parameter value θ that maximizes the likelihood function $L(\theta) = P(\text{data}|\theta)$ — i.e., the value that makes the observed data most probable. Steps: (1) Write likelihood function L(θ); (2) Take log-likelihood: ln L(θ); (3) Differentiate w.r.t. θ and set = 0; (4) Solve for $\hat{\theta}_{MLE}$. Example (Bernoulli): Coin tosses H,H,T,H,T → $\hat{\theta}_{MLE} = 3/5 = 0.6$. Example (Binomial): n=10, x=2 defectives → $\hat{p}_{MLE} = x/n = 0.2$. Advantages: Consistent, asymptotically efficient, invariance property. Limitations: Biased in small samples; computationally intensive for complex models.

45. [IAE QB-1 | B3] z-test: A manufacturer claims batteries last μ=50h. Test at 5% significance: n=36, $\bar{x}$=53h, σ=9h — state H_0/H_1, compute test statistic, and give conclusion *(Numerical)*

Answer: H_0: $\mu = 50$h (manufacturer's claim is true). H_1: $\mu \neq 50$h (two-tailed — testing if batteries last a different amount). Formula: $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n}) = (53 - 50)/(9/\sqrt{36}) = 3/(9/6) = 3/1.5 = 2.0$. At α=0.05 (two-tailed), z-critical = ±1.96. Since $|2.0| > 1.96$, REJECT H_0. Conclusion: There is sufficient statistical evidence at 5% significance level to reject the manufacturer's claim. The batteries do not last exactly 50 hours.

Module III Regression Models, Regularization & Nonlinear Models

Syllabus Topics: *Simple & multiple linear regression · LLSR derivation · Model assumptions · Residual analysis · Multicollinearity (VIF) · Polynomial & interaction effects · Nonlinear regression · Regularization: Ridge, Lasso, Elastic Net · Spline regression & LOESS · GLM: Logistic regression (binary), Poisson (count data) · Model evaluation: MAE, MSE, RMSE, R^2, Adjusted R^2*

Total: 13 questions · Dec 2022: 2 · Dec 2023: 2 · May 2024: 1 · Dec 2024: 1 · May 2025 (AIML): 2 · May 2025 (IT): 4 · IAE QB-1: 1

1. **[Dec 2022 | Q4b]** Multiple regression: find relation between GPA of students and number of study hours and height *(Numerical)*

2. **[Dec 2022 | Q5b]** Simple linear regression equation + coefficient of determination for Hour–Temperature data *(Numerical)*

3. **[Dec 2023 | Q1c]** Short note on Simple Linear Regression *(Short note)*

4. **[Dec 2023 | Q5b]** Explain Linear Least Square Regression (LLSR) with advantages and disadvantages *(Theory)*

5. **[May 2024 | Q5b]** Find simple linear regression equation and coefficient of determination (R^2) for given X–Y data *(Numerical)*

6. **[Dec 2024 | Q3a]** Explain single and Multiple linear regression with example and suitable plot *(Theory)*

7. **[May 2025 (AIML) | Q1d]** Explain Linear Regression and its Applications *(Short note)*

8. **[May 2025 (AIML) | Q5b]** Find the simple linear regression equation for Time–Growth data *(Numerical)*

9. **[May 2025 (IT) | Q3b]** Differentiate between Simple Linear Regression and Multiple Regression *(Theory)*

10. **[May 2025 (IT) | Q4a]** What is Logistic Regression? *(Theory)*

11. **[May 2025 (IT) | Q4b]** Find simple linear regression equation for Bill–Tip data *(Numerical)*

12. **[May 2025 (IT) | Q5b]** Derive Simple Linear Regression parameters to predict the value of the output *(Derivation)*

13. **[IAE QB-1 | A10]** What is multicollinearity in regression? Explain its effect on the model *(Theory)*

Module IV Classification & Supervised Learning

Syllabus Topics: *Classification metrics (accuracy, precision, recall, F1, ROC-AUC, PR curves) · Logistic regression, odds ratio · Naive Bayes · k-NN classifier · Decision Trees & pruning · Ensemble methods: Bagging, Random Forest, Gradient Boosting · SVM (linear & kernel) · Handling imbalanced data: SMOTE · Confusion matrix analysis*

Total: 7 questions · IAE QB-2: 7

1. [IAE QB-2 | A1] What is the difference between precision and recall? Mention one situation where recall is preferred over precision *(Theory)*

2. [IAE QB-2 | A2] A classification model produces TP=40, FP=10, FN=5, TN=45. Calculate the precision of the model *(Numerical)*

3. [IAE QB-2 | A3] What is SMOTE? Why is it used when handling an imbalanced classification dataset? *(Theory)*

4. [IAE QB-2 | A4] What is the kernel trick in SVM? State its purpose *(Theory)*

5. [IAE QB-2 | B1] Classification model for fraudulent transactions (imbalanced dataset): a) Why can accuracy be misleading? b) How do precision, recall, and F1-score provide better information? c) Which metric to prioritize to detect as many fraudulent transactions as possible? *(Theory)*

6. [IAE QB-2 | B2] A Decision Tree gives 95% training accuracy but 78% testing accuracy. a) Identify the likely problem. b) Explain how pruning can help. c) State one advantage of Random Forest over a single Decision Tree *(Theory)*

7. [IAE QB-2 | B3] Explain the working principle of SVM: a) What is a hyperplane? b) What is the margin and why is maximizing it important? c) Explain the kernel trick with one suitable example *(Theory)*

Note: No questions from PYQs map to this module. All 7 questions above are from the IAE Question Bank (QB-2) only. Classification algorithms remain high-priority for upcoming exams.

Module V Unsupervised Learning & Dimensionality Reduction

Syllabus Topics: *PCA (covariance matrix, Eigen decomposition, explained variance) · LDA (supervised dim. reduction) · Kernel PCA & ICA (conceptual) · Clustering: K-means, Hierarchical, DBSCAN · Cluster evaluation: silhouette score, Davies-Bouldin index · Dimensionality reduction: t-SNE, UMAP · Outlier detection: Z-score, IQR, Isolation Forest, LOF*

Total: 8 questions · Dec 2024: 1 · IAE QB-2: 7

1. [Dec 2024 | Q1e] How to detect outliers? *(Short note)*

2. [IAE QB-2 | A5] What is the purpose of explained variance in PCA? How does it help in selecting the number of principal components? *(Theory)*

3. [IAE QB-2 | A6] Mean of feature=50, SD=5. Calculate Z-score for observation value=65. Is it an outlier using |Z|>3 criterion? *(Numerical)*

4. [IAE QB-2 | A7] What is the main difference between PCA and LDA in terms of whether they use class labels? *(Theory)*

5. [IAE QB-2 | A8] What is the silhouette score in clustering? What does a high silhouette score indicate about cluster quality? *(Theory)*

6. [IAE QB-2 | B4] PCA on correlated features: a) Explain how PCA uses covariance matrix and eigen decomposition to reduce dimensionality. b) What is explained variance? c) Why is retaining the first few principal components useful for ML models? *(Theory)*

7. [IAE QB-2 | B5] Customer transaction dataset: a) Explain K-means clustering for identifying customer groups. b) State one limitation of K-means and how DBSCAN addresses it. c) Explain how silhouette score evaluates cluster quality *(Theory)*

8. [IAE QB-2 | B6] High-dimensional data with potential outliers: a) Explain purpose of t-SNE or UMAP for visualizing high-dimensional data. b) Explain IQR method for outlier detection. c) Name one tree-based method for outlier detection and briefly explain its idea *(Theory)*

Note: Only 1 question from PYQs (outlier detection). 7 additional questions are from the IAE Question Bank (QB-2). PCA, K-means, DBSCAN, t-SNE, and silhouette score are high-potential exam topics.

Module VI Model Evaluation, Bayesian Inference & Ethics

Syllabus Topics: *Model selection: AIC, BIC · Bias-variance trade-off · Cross-validation: k-fold, LOOCV · Calibration curves & Brier score · Bayesian fundamentals: prior, likelihood, posterior · MAP vs MLE · Conjugate priors · Naive Bayes from Bayesian view · Interpretability: SHAP, LIME, PDP, ICE · Fairness in ML · Ethical considerations · Data privacy · Reproducible ML (MLflow, W&B)*

Total: 3 questions · May 2025 (IT): 1 · IAE QB-2: 2

1. [May 2025 (IT) | Q1a] What do you mean by random variables? Explain Bayes theorem *(Theory)*

2. [IAE QB-2 | A9] Differentiate between AIC and BIC as model selection criteria. Mention one key difference in how they penalize model complexity *(Theory)*

3. [IAE QB-2 | A10] A model has high training accuracy but significantly lower validation accuracy. Identify the likely issue using the bias–variance trade-off *(Theory)*

Note: Only 1 question from PYQs (Bayes theorem). 2 additional questions are from the IAE Question Bank (QB-2). AIC/BIC and bias-variance trade-off are now covered via IAE QB. Cross-validation, SHAP/LIME, and ML fairness remain untested.

KEY TAKEAWAYS FOR EXAM PREP

1. **Modules I & II dominate every paper (~65–70% of marks)**
   Every paper has hypothesis testing questions (z-test, t-test, ANOVA, Chi-square), normal/Poisson distribution numericals, confidence intervals, and descriptive statistics. Prioritise these two modules above all others.
2. **Module III (Regression) appears in every paper**
   Expect at least 1–2 numerical (find the SLR equation for given data) and 1 theory question (explain LLSR / multiple regression). Know how to compute regression coefficients, R^2, and residuals.
3. **Hypothesis testing numericals are repeated every exam**
   z-test, t-test, one-way ANOVA, and Chi-square independence tests have numerical problems in every paper. Memorise the formulas: test statistic, degrees of freedom, and decision rule.
4. **Nonparametric tests are increasing (Dec 2024 → May 2025)**
   Friedman test, Kruskal-Wallis test, and Wilcoxon Signed-Rank test appeared in the newer papers. Prepare these as short-note or theory questions, not just numericals.
5. **Modules IV, V, VI: IAE QB covers what PYQs miss**
   Classification (SVM, Decision Trees, Random Forest, SMOTE), PCA/clustering (K-means, DBSCAN, silhouette score), and model evaluation (AIC, BIC, bias-variance) are covered in the IAE Question Bank but absent from PYQs. Prepare SVM working principle, PCA with eigen decomposition, and confusion-matrix numericals as priority topics.
6. **Recurring short-note topics across papers**
   Histogram, Box Plot, Stem & Leaf, Weibull distribution, Type I/II errors, Fisher's exact test, and Bootstrapping appear as short notes in multiple papers. Prepare 5-line answers for each.